Visual Analytics Project Proposal

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1 General idea

We are proposing to develop a visualization system with the purpose of providing general insights on the topic of vehicle accidents in Italy.

Intended users

The intended users for this system would be members of the Ministry of Transport, police and traffic authorities, helping them with taking informed decision on how to better prevent accidents in Italy, particularly those involving injuries.

2 Dataset

The dataset contains details about vehicle accidents with injuries occurred in Italy during 2024. This dataset was extracted from the data published on the ISTAT website and comprises about 92.000 tuples, each described by 8 attributes.

The attributes in the dataset are in summary: region and time (hour, weekday, month) of accident, road section where the accident occurred (traffic circle, intersection, etc...), type of accident (between vehicles, vehicle-pedestrian, etc...), deadliness of the accident, and number of accidents registered with the given characteristics.

More details are provided with a table at the end of the proposal.

3 Visualization System Description

The system is composed of two main sections as shown in Figure 1.

Section A visualizes the number of accidents for each main attribute of the dataset. This is is done through different bar charts with categorical values displayed along the horizontal axis (X) and the number of accidents along the vertical axis (Y). These charts are coordinated with each other. As an example, selecting the accidents that occurred between January and March also highlights the corresponding road section types and accident categories within that timeframe.

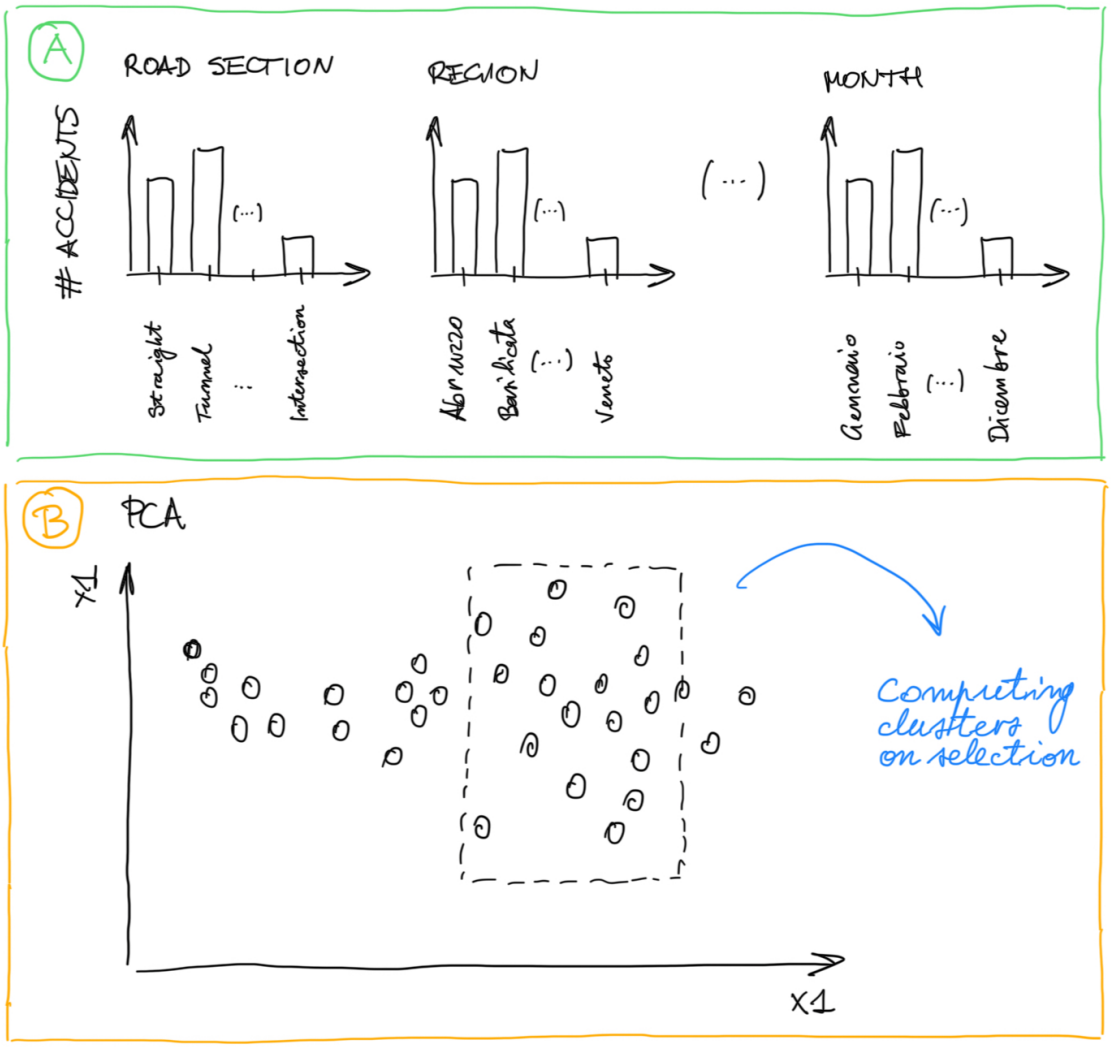

Figure 1: Draft of the visualization system.

Section B presents the result of a PCA performed on the dataset. Data points can be colored according to the categorical attributes of the dataset: accident type, deadliness etc. The PCA graph is coordinated with the charts in section A. As an example, selecting accidents that occurred between January and March in Section A highlights the corresponding data points in the PCA plot, and vice versa.

Furthermore, the user can visually trigger an analytic directly brushing on the PCA graph, the system will respond by searching for clusters (*e.g. with k-means*) within the selected subset of data points.

4 Dataset Details

Attribute	Description
Area	ID of Region
Road section	Number identifying a road section. Identifies: tunnel, bend, crossroad, traffic circle, bump/slope or bottleneck, straight stretch, level crossing.
Accident type	Number identifying the accident type. Identifies: accidents between vehicles, vehicle-pedestrian accidents and accidents involving a single vehicle.
Deadliness	Boolean attribute that identifies if the accident involved death or only injury
Hour of accident	Time of accident measured with hour resolution
Month	Month number
Weekday	Number identifying the weekdays: monday, tuesday, wednesday, etc...
Observations	Number of accidents occurred with the specified attributes